Local Linear Spatial–Spectral Probabilistic Distribution for Hyperspectral Image Classification

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***Abstract*— A key challenge in hyperspectral image (HSI) classification is how to effectively utilize the spectral and spatial information of limited labeled training samples in the data set. In this article, a new spatial–spectral combined classification method, termed local linear spatial-spectral probabilistic distribution (LSPD), has been proposed on the basis of local geometric structure and spatial consistency of HSI. LSPD extracts discriminating spatial–spectral information from limited labeled training samples and their spatial–spectral neighbors. Then, it constructs a multiclass probability map by exploiting the local linear representation and spatial information of HSI. Finally, the spatial–spectral weighted reconstruction has been performed on the probability map, and the class of test sample can be predicted by the maximum value of LSPD. LSPD not only exploits spectral information to discover more intrinsic properties of the labeled training data but also utilizes the spatial relationship between samples to effectively improve discriminating power for classification. Experimental results on the Indian Pines, PaviaU, and HoustonU hyperspectral data sets demonstrate that the proposed LSPD method possesses better classification performance by comparing with some state-of-the-art classifiers.**

***Index Terms*— Classification, hyperspectral image (HSI), probability map, spatial information, spatial–spectral reconstruction.**

I. Introduction

HYPERSPECTRAL image (HSI) as a kind of remote sensing image records dozens to hundreds of radiation spectrum bands of the earth surface. It provides rich spectral features and spatial information for classifying land-cover types [1], [2]. Therefore, HSI has been widely applied in various fields, such as precision agriculture, earth observation, environmental monitoring, and mineral detection [3]. In these applications, how to exploit the abundant information in HSI for accurately classifying ground objects is a key issue that has attracted increasing attention in recent years [4].

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Many HSI classification methods have been developed based on the abundant spectral information in HSI, in which k-nearest neighbor (KNN) [5], spectral angle mapper (SAM) [6], support vector machine (SVM) [7], and random forests (RF) [8] are some common classifiers applied in HSI classification. Although these algorithms can infer the labels of samples via spectral similarities between pixels, they ignore the intrinsic characteristics of HSI data space, such as geometric structure and sparse property [9], [10]. To address this problem, Fu *et al*. [11] designed a classification algorithm based on the concept of KNN simplex (KNNS), which determines the class label of test sample according to the linear reconstruction errors on different classes. Li *et al*. [12] presented a nearest-subspace (NRS) classifier that seeks an approximation of each test sample via linear combination within each class, and the class label is derived according to the class, which best approximates the test sample. Wright *et al*. [13] proposed sparse representation-based classification (SRC) relied on computing sparse reconstruction errors of test samples with dictionaries from different classes. These classifiers utilize the intrinsic properties in the spectral domain of HSI to achieve good classification results.

However, the aforementioned classifiers are pixel-wise methods, which only utilize the spectral features of pixels but abandon the valuable spatial information in HSI [14]. For HSI, pixels within a small spatial patch usually have similar spectral signatures, and they probably come from the same class [15], [16]. Recent investigations show that incorporating spatial information into traditional spectral-based classifiers can further improve the performance of HSI classification [17]. In [18] and [19], the image patches distance (IPD) and the spatial coherence distance (SCD) were put forward to replace the Euclidean distance in KNN, and the modified IPD-KNN and SCD-KNN classifiers achieved higher classification accuracies than KNN. Extended multivariate attribute profile (EMAP), multiscale-independent component analysis (MICA), and gray-level co-occurrence matrices (GLCM) are three popular spatial feature learning approaches for HSI data [20]–[22]. These methods respectively extract the extend multilevel spatial features, the low-level feature representation, and the spatial texture features of HSI, and then, they exploit SVM to classify these spatial features. Using a spatial filter is also a common strategy for HSI classification. Kang *et al*. [23], [24] put forward two filters called edge-preserving filtering (EPF) and image fusion and recursive

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filtering (IFRF) to obtain the spatial filtered features for improving classification performance. In [25], the presented SVMCK method exploited composite kernels (CKs) to compute spatial–spectral weighted mean features and achieved better classification results than SVM.

Compared with the exploiting spatial–spectral distance or spatial–spectral mean features in traditional classifiers, another classification strategy is to utilize spatial correlation via a representation model or Markov random field (MRF). By means of a sparse representation (SR) model, Chen *et al*. [26] presented two jointly spatial–spectral SR classifiers termed simultaneously orthogonal matching pursuit (SOMP) and simultaneously subspace pursuit (SSP), which computed local spatial simultaneous sparse coefficients based on the algorithms OMP and SP. Li and Du [27] incorporated local spatial information into a collaborative representation (CR) model and a joint CR (JCR) classifier for HSI.

Besides the SR and CR models, MRF is an efficient tool to fuse spatial prior information for improving the spatial smoothness and classification accuracy of HSI. Sparse representation-based MRF (SRMRF) and SVM-based MRF (SVMMRF) are the two spatial–spectral classifiers based on MRF [28], [29]. They first compute the class probability of each sample with SR or SVM and then use the probability knowledge and spatial information of samples to construct the MRF model for classification. In addition, Pan *et al*. [30] presented a MRF-based method aICPssi by integrating adaptive interclass-pair penalty (aICP) and spectral similarity information (SSI) of HSI. Sun *et al*. [31] combined the weighted MRF model and spatially adaptive total variation regularization to form a spatial smooth classifier named SMLR-SpATV for HSI classification.

In recent years, deep learning has received considerable attention in hyperspectral remote sensing field, and some deep network models have been developed for HSI classification, such as deep belief network (DBN) [32], convolutional neural networks (CNNs) [33], stacked convolutional autoencoder (SCAE) [21], and stacked sparse autoencoder model (SSAE) [34]. In addition, some new deep networks were proposed based on the spatial and spectral characteristics of HSI. To utilize spatial and spectral information simultaneously, researchers put forward a spatial–spectral DBN (SSDBN) model that stacks spatial and spectral feature vectors in the DBN to learn the labels of samples [35]. In [36], a self-taught semisupervised autoencoder (SuSA) was proposed on the basis of a multiloss convolutional autoencoder and a semisupervised multilayer perceptron, in which the former model extracts the spatial–spectral features and the latter model classifies the features of samples. These spatial–spectral modified networks achieve satisfying classification results in comparison with the traditional deep learning algorithms.

However, these deep learning methods usually require a great number of labeled samples to guarantee the training process, but the labeled samples in HSIs are usually difficult or time-consuming to obtain, which limits the application of deep learning methods in real HSI classification tasks. For the aforementioned representation-based and MRF-based methods, they only utilize the spatial locations of pixels but ignore the discriminant spatial relationship between the labeled and unlabeled samples [37]. Thus, they abandon rich discriminant spatial–spectral information in training processing that results in unsatisfying classification performance with the small labeled training set. In addition, these algorithms do not take into account the intrinsic geometric structure in the HSI data space, so the improvement of classification performance has been further restricted.

To overcome the aforementioned drawbacks, a novel spatial–spectral combined classification approach termed local linear spatial–spectral probabilistic distribution (LSPD) has been proposed for HSI classification. At first, LSPD exploits labeled samples and their spatial–spectral neighbors to construct a spatial–spectral training set and provide more discriminant spatial–spectral information for classification. Then, LSPD obtains the probability of test pixels belonging to each class by calculating local weighted linear representation errors, and it constructs a multiclass probability map according to the spatial coordinates of pixels. Finally, LSPD conducts a spatial–spectral reconstruction on the multiclass probability map to obtain the spatial–spectral probabilistic distributions and predict the labels of test samples. The main contributions of this article are listed as follows.

1) Based on the phenomenon that pixels in a small spatial region of HSI are more likely to form the same class, the proposed LSPD designs a new neighbor selection model to obtain the spatial–spectral neighbors of labeled samples for training. The constructed spatial–spectral training set contains more discriminant information extracted from the spatial and spectral domains of HSI, which is beneficial to improve the classification performance with limited labeled training samples.
2) Samples from the same class generally possess similar geometric structure in data space, and intraclass samples can be accurately represented by each other. LSPD linearly represents each test sample with different class training sets and calculates the class probability of a test sample by means of minimizing local weighted linear representation error, which fully exploits the property of data distribution to enhance the classification accuracy.
3) LSPD constructs a multiclass probability map and performs spatial–spectral weighted reconstruction on the map to obtain the spatial–spectral probabilistic distribution of the test sample. This procedure effectively enhances the aggregation between the homogeneous data and the separability of data from different classes, which is helpful to improve the discriminating power for HSI classification.

The remainder of this article is organized as follows. In Section II, the proposed LSPD method is introduced in detail. Experiments on some HSI data sets have been presented in Section III. Finally, Section IV summarizes this article and provides some suggestions for future work.

II. Local Linear Spatial–Spectral Probabilistic Distribution Classification Method

Based on the spatial and spectral correlation in HSI, a novel classification method, termed LSPD, has been proposed in

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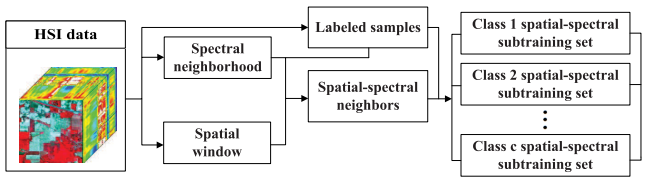


Fig. 1. Procedure of LSPD selecting spatial–spectral neighbors for training.

this article. LSPD designs a new neighbor selection model for building the spatial–spectral training set that contains more discriminant information, and it exploits the local discriminant geometric structure to construct a local linear probabilistic distribution model for classification. LSPD fully utilizes the spatial and spectral information in HSI and significantly enhances the classification performance for HSI.

A. Spatial–Spectral Training Set

The land-cover types in HSI usually present aggregation distribution because the same ground objects, such as building, road, rivers, and trees, generally exist in the continuous regions. Thus, the pixels in a small spatial region are more likely to belong to the same class, which reflects the correlation between the pixels in the spatial domain. Based on this spatial correlation, the proposed LSPD tries to select the spatial–spectral neighbors of labeled samples and add them into a training set for classification. Due to the spatial–spectral neighbors of labeled samples that usually have the same labels with center samples, the constructed spatial–spectral training set containing more discriminant information can effectively address the problem of insufficient labeled samples. The procedure of LSPD selecting spatial–spectral neighbors for training is shown in Fig. 1.

The effectiveness of spatial–spectral training set is mainly depended on the way of selecting the neighbors. To select appropriate neighbors, we design a new spatial–spectral neighbor selection model. For convenience, we first define an HSI data set as $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n] \in \Re^{D\times n}$, in which D and n are the numbers of spatial bands and pixels in HSI, respectively. $l_i \in \{1, 2, \ldots, c\}$ represents the class label of pixel $\mathbf{x}_i$, and c is the number of classes in HSI.

In the neighbor selection model, a $w \times w$ square spatial window of center pixel $\mathbf{x}_i(p_i, q_i)$ is applied to select spatial neighbor set, where w is an odd positive integer and (p_i, q_i) denotes the coordinate of the pixel $\mathbf{x}_i$. The spatial neighbor set $\mathbf{\Omega}_{\text{spatial}}(\mathbf{x}_i)$ is defined as

$$\mathbf{\Omega}_{\text{spatial}}(\mathbf{x}_i) = \left\{ \mathbf{x}(p,q) \,\middle|\, \begin{array}{l} p \in [p_i - a, p_i + a] \\ q \in [q_i - a, q_i + a] \end{array} \right\} \quad (1)$$

in which $a = (w-1)/2$. Then, $w^2 - 1$ spectral nearest neighbors of $\mathbf{x}_i$ are chosen on the basis of the spectral Euclidean distance, and the spectral neighbor set $\mathbf{\Omega}_{\text{spectial}}(\mathbf{x}_i)$ is built by

$$\mathbf{\Omega}_{\text{spectial}}(\mathbf{x}_i) = \left\{ \mathbf{x}_i^1, \mathbf{x}_i^2, \ldots, \mathbf{x}_i^n \right\}, \quad n = 1, \ldots, w^2 - 1 \quad (2)$$

where $\mathbf{x}_i^n$ is the nth spectral nearest neighbor of $\mathbf{x}_i$. Based on the above-mentioned spatial and spectral neighbors, the final spatial–spectral neighbor set $\mathbf{\Omega}_{\text{ss}}(\mathbf{x}_i)$ is constructed as

$$\mathbf{\Omega}_{\text{ss}}(\mathbf{x}_i) = \{\mathbf{x} | \mathbf{x} \in \mathbf{\Omega}_{\text{spatial}}(\mathbf{x}_i) \cap \mathbf{x} \in \mathbf{\Omega}_{\text{spectial}}(\mathbf{x}_i)\} \quad (3)$$

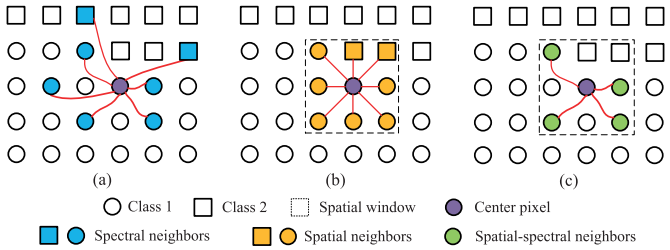


Fig. 2. Different neighbors of samples in HSI. (a) Spectral neighbors. (b) Spatial neighbors. (c) Spatial–spectral neighbors.

Algorithm 1 Spatial–Spectral Training Set

Input: Labeled training set $\mathbf{X}^L = [\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_m]$ with labels $\{l_1, l_2, \cdots, l_m\}$, the size of spatial window w, initial spatial-spectral training set $\mathbf{X}^S = \{\mathbf{X}_1^S, \ldots, \mathbf{X}_c^S\} = \emptyset$.

1: **for** $i = 1 : m$
2: Ensure the label of $\mathbf{x}_i$, $l = l_i$;
3: Select the spatial neighbor set $\mathbf{\Omega}_{spatial}(\mathbf{x}_i)$ by (1);
4: Find the spectral neighbor set $\mathbf{\Omega}_{spectral}(\mathbf{x}_i)$ by (2);
5: Build the spatial-spectral neighbor set $\mathbf{\Omega}_{ss}(\mathbf{x}_i)$ by (3);
6: Add $\mathbf{\Omega}_{ss}(\mathbf{x}_i)$ into l-th class sub-training set $\mathbf{X}_l^S$;
7: **end**

Output: Spatial-spectral training set: $\mathbf{X}^S = \{\mathbf{X}_1^S, \ldots, \mathbf{X}_c^S\}$.

in which the neighbors are selected from the intersection of $\mathbf{\Omega}_{\text{spatial}}(\mathbf{x}_i)$ and $\mathbf{\Omega}_{\text{spectial}}(\mathbf{x}_i)$.

To illustrate the merits of spatial–spectral neighbors, Fig. 2 shows the traditional spectral neighbors, spatial neighbors, and the proposed spatial–spectral neighbors. As shown in Fig. 2(a), seven neighbors of the center pixel are selected in the light of spectral similarity. Due to the fact that different objects in HSI may have similar spectrums, spectral neighbors may be chosen from the classes different from the center pixel. In Fig. 2(b), all pixels in a spatial window are chosen as the neighbors of center pixel. However, the spatial window may contain pixels from different classes, especially when the center pixel locates on the border of different ground objects. For spatial–spectral neighbors in Fig. 2(c), LSPD selects the neighbors simultaneously belonging to the spatial and spectral neighbors of the center pixel. If spectral neighbors of one pixel all located outside spatial window, the pixel will be considered as a noise point because it has a large spectral difference with all around pixels, and the neighbors of this pixel will also not be selected for training. Therefore, the proposed spatial–spectral neighbor selection effectively integrates both spectral and spatial similarities for ensuring a training set that contains more useful spatial–spectral discriminant information.

Based on the designed neighbors selection model, LSPD first chooses the spatial–spectral neighbors of each labeled sample, and then, it exploits the limited labeled samples with their neighbors to construct a spatial–spectral training set $\mathbf{X}^S = \{\mathbf{X}_l{}^S : l \in \{1, 2, \ldots, c\}\}$, in which $\mathbf{X}_l^S = \{\mathbf{x}_1^l, \mathbf{x}_2^l, \ldots, \mathbf{x}_{N_l}^l\}$ is a subtraining set that contains N_l samples of lth class. The detailed steps of constructing spatial–spectral training set are shown in Algorithm 1.

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B. Local Linear Spatial–Spectral Probabilistic Distribution

Under the assumption that pixels from the same class usually possess similar reflection spectrums, an unknown sample can be accurately represented by a linear combination of intraclass training samples. On the contrary, if an unknown sample is linearly represented by interclass training samples, the corresponding linear representation result will deviate from the true value. The linear representation error can reflect the probability of an unknown pixel belonging to one specific class. Therefore, the probabilities of pixels belonging to different classes can be calculated by linear representation errors, and the label of an unknown pixel will be determined by comparing its class probabilities.

Suppose $\mathbf{x}_t$ is a test sample, and the number of samples in a subtraining set $\mathbf{X}_l^S = \{\mathbf{x}_1^l, \mathbf{x}_2^l, \ldots, \mathbf{x}_{N_l}^l\}$ is N_l. $\mathbf{x}_t$ can be linearly represented by the lth class training samples as

$$\mathbf{x}_t \approx v_1^l \mathbf{x}_1^l + v_2^l \mathbf{x}_2^l + \cdots + v_{N_l}^l \mathbf{x}_{N_l}^l \tag{4}$$

in which v_i^l is the weight of training sample $\mathbf{x}_t^l$ in $\mathbf{X}_l{}^S$, and it controls the contribution of $\mathbf{x}_t^l$ in a linear representation model.

In real HSI applications, the linear representation using training samples from the same class still probably produces larger error due to the phenomenon that the same objects may have different spectrums. To avoid this problem, we select KNNs from each class for linear representation. The KNN training set $\{\mathbf{x}_{tj}^l\}_{j=1,\ldots,k}$ of $\mathbf{x}_t$ is selected from the lth class, and the test sample $\mathbf{x}_t$ can be represented by a linear combination of the selected neighbors as follows:

$$\begin{aligned}\mathbf{x}_t &\approx v_{t1}^l \mathbf{x}_{t1}^l + v_{t2}^l \mathbf{x}_{t2}^l + \cdots + v_{tk}^l \mathbf{x}_{tk}^l \\ &= \left[\mathbf{x}_{t1}^l, \mathbf{x}_{t2}^l, \ldots, \mathbf{x}_{tk}^l\right]\left[v_{t1}^l, v_{t2}^l, \ldots, v_{tk}^l\right]^T \\ &= \mathbf{K}_t^l \mathbf{v}_t^l \end{aligned} \tag{5}$$

where $\mathbf{K}_t^l$ is a $D \times k$ matrix termed KNN dictionary whose columns are training samples and $\mathbf{v}_t^l$ is a k-dimensional vector whose entries are the weights of the corresponding atoms in $\mathbf{K}_t^l$.

To obtain normalization weights, a constraint $\mathbf{I}^T \mathbf{v}_t^l = 1$ is applied into an optimization model, and the weight vector $\mathbf{v}_t^l$ is computed by minimizing the linear representation error as

$$\begin{cases} J\left(\mathbf{v}_t^l\right) = \min \left\|\mathbf{x}_t - \mathbf{K}_t^l \mathbf{v}_t^l\right\|^2 \\ \text{s.t. } \mathbf{I}^T \mathbf{v}_t^l = 1 \end{cases} \tag{6}$$

where $\mathbf{I}^T = [1, 1, \ldots, 1]_{1\times k}$.

With some mathematical operations, the linear representation error can be reduced to

$$\begin{aligned}&\left\|\mathbf{x}_t - \mathbf{K}_t^l \mathbf{v}_t^l\right\|^2 \\ &= \left\|\mathbf{x}_t - \sum_{j=1}^{k} v_{tj}^l \mathbf{x}_{tj}^l\right\|^2 \\ &= \left\|\mathbf{x}_t \sum_{j=1}^{k} v_{tj}^l - \sum_{j=1}^{k} v_{tj}^l \mathbf{x}_{tj}^l\right\|^2 \\ &= \left\|\begin{bmatrix}\mathbf{x}_t - \mathbf{x}_{t1}^l & \mathbf{x}_t - \mathbf{x}_{t2}^l & \cdots & \mathbf{x}_t - \mathbf{x}_{tk}^l\end{bmatrix}\mathbf{v}_t^l\right\|^2 \\ &= \left(\mathbf{v}_t^l\right)^T \mathbf{Z}_t^l \mathbf{v}_t^l. \end{aligned} \tag{7}$$

Then, (6) can be denoted as

$$\begin{cases} J\left(\mathbf{v}_t^l\right) = \min \left(\mathbf{v}_t^l\right)^T \mathbf{Z}_t^l \mathbf{v}_t^l \\ \text{s.t. } \left(\mathbf{v}_t^l\right)^T \mathbf{I} = 1. \end{cases} \tag{8}$$

According to the Lagrange multiplier method, the optimization solution of (8) is obtained as follows:

$$\mathbf{v}_t^l = \frac{\lambda\left(\mathbf{Z}_t^l\right)^{-1}\mathbf{I}}{2}. \tag{9}$$

To meet the constraint condition in an optimization model, the weights vector $\mathbf{v}_i^l$ can be normalized by

$$\mathbf{v}_t^l = \frac{\frac{\lambda\left(\mathbf{Z}_t^l\right)^{-1}\mathbf{I}}{2}}{\frac{\lambda \mathbf{I}^T\left(\mathbf{Z}_t^l\right)^{-1}\mathbf{I}}{2}} = \frac{\left(\mathbf{Z}_t^l\right)^{-1}\mathbf{I}}{\mathbf{I}^T\left(\mathbf{Z}_t^l\right)^{-1}\mathbf{I}} \tag{10}$$

and the test sample $\mathbf{x}_t$ can be linearly represented by $\mathbf{K}_t^l \mathbf{v}_t^l$.

With the linear representation of $\mathbf{x}_t$, the probability of $\mathbf{x}_t$ belonging to each class can be calculated through the following three major steps: 1) building KNN dictionaries $\{\mathbf{K}_t^l : l \in \{1, 2, \ldots, c\}\}$ from c classes; 2) computing the weight vectors $\{\mathbf{v}_t^l : l \in \{1, 2, \ldots, c\}\}$ of $\mathbf{x}_t$; and 3) calculating the probabilities $\{p_t^l : l \in \{1, 2, \ldots, c\}\}$, in which p_t^l is defined as:

$$p_t^l = \frac{\left\|\mathbf{x}_t - \mathbf{K}_t^l \mathbf{v}_t^l\right\|^{-1}}{\sum_{j=1}^{c}\left\|\mathbf{x}_t - \mathbf{K}_t^j \mathbf{v}_t^j\right\|^{-1}} \tag{11}$$

where p_t^l denotes the probability of $\mathbf{x}_t$ belonging to the lth class.

To further utilize spatial information for classification, the class probabilities of test samples are exploited to construct a multiclass probability map according to the corresponding spatial coordinates. Based on the spatial relationship between pixels, the probability map is spatial–spectral reconstructed to obtain the probabilistic distribution of test samples. Fig. 3 shows the steps of processing a multiclass probability map.

As shown in Fig. 3, the probabilities $\{p_t^l : l \in \{1, 2, \ldots, c\}\}$ have been vectored as the probabilistic vector $\boldsymbol{\beta}_{t(p_t,q_t)}$ representing the probability distribution of $\mathbf{x}_t$. Then, the multiclass probability map is developed with the probabilistic vectors and the corresponding coordinates, in which the ith layer contains the probabilities of pixels belonging to the ith class.

To make full use of the spatial information in a multiclass probability map, a spatial–spectral weighted reconstruction is conducted on the map as

$$\begin{cases} \hat{\boldsymbol{\beta}}_t = \displaystyle\sum_{\boldsymbol{\beta}_{tk} \in \boldsymbol{\Omega}(\boldsymbol{\beta}_t)} b_{tk}\boldsymbol{\beta}_{tk} = \boldsymbol{\beta}_t + \sum_{k=1}^{w^2-1} b_{tk}\boldsymbol{\beta}_{tk} \\ b_{tk} = \exp\{-\alpha\|\boldsymbol{\beta}_t - \boldsymbol{\beta}_{tk}\|^2\} \end{cases} \tag{12}$$

where spatial neighborhood $\boldsymbol{\Omega}_{\text{spatial}}(\boldsymbol{\beta}_t)$ of $\boldsymbol{\beta}_t$ is selected according to (1). $\boldsymbol{\beta}_{tk}$ is the probabilistic vector of kth neighbor in spatial neighborhood, b_{tk} is the reconstruction weight between $\boldsymbol{\beta}_t$ and $\boldsymbol{\beta}_{tk}$, and tradeoff parameter α controls the influence of distance between the probabilistic vectors.

After spatial–spectral weighted reconstruction, we obtain the probabilistic vector $\hat{\boldsymbol{\beta}}_t = [\hat{p}_t^1, \hat{p}_t^2, \ldots, \hat{p}_t^c]^T$ that represents

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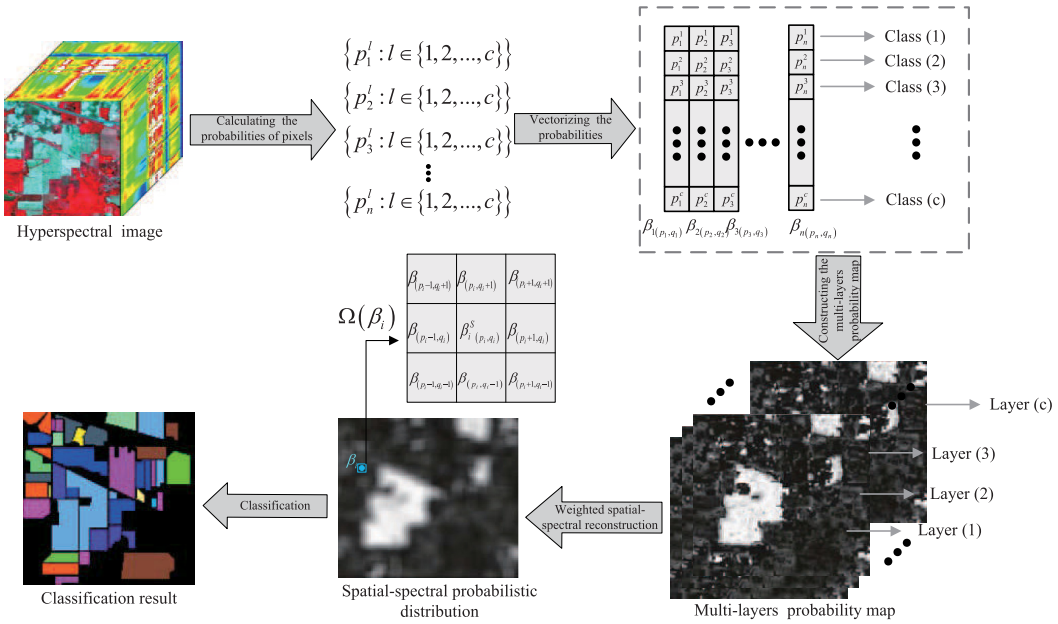


Fig. 3. Steps of LSPD for classification.

Algorithm 2 LSPD

Input: Hyperspectral image, spatial-spectral training data set $\mathbf{X}^S = \{\mathbf{X}_1^S, \ldots, \mathbf{X}_c^S\}$, the size of spatial window w, the number of spectral neighbors k, reconstruction parameter t, and test sample $\mathbf{x}_t$.

1: Construct k-nearest-neighbor dictionaries $\{\mathbf{K}_t^l : l \in \{1, 2, \cdots, c\}\}$ from c classes subtraining sets $\{\mathbf{X}_l^S : l \in \{1, 2, \cdots, c\}\}$;
2: **for** $i = 1 : c$
3: Compute the weight vectors $\mathbf{v}_t^l$ of $\mathbf{x}_t$ in i-th class dictionary $\mathbf{K}_t^l$ by Eqs. (8)-(10);
4: Calculate the i-th class probability p_t^l of $\mathbf{x}_t$ by Eq. (11);
5: **end**
6: Obtain the probabilistic vector $\boldsymbol{\beta}_t = \left[p_t^1, p_t^2, \ldots, p_t^c\right]^T$;
7: Construct multiclass probability map and find the spatial neighbors $\boldsymbol{\Omega}_{spatial}\left(\boldsymbol{\beta}_t\right)$ by Eq. (1);
8: Compute the spatial-spectral probabilistic vector $\hat{\boldsymbol{\beta}}_t$ by Eq. (12);
9: Infer the label l_t^* of $\mathbf{x}_t$ with Eq. (13);

Output: class label:l_t^*

the LSPD of $\mathbf{x}_t$. The element $\hat{p}_t^c$ denotes the spatial–spectral probability of test sample $\mathbf{x}_t$ belonging to class c. Finally, the label l_t^* of $\mathbf{x}_t$ can be inferred with

$$l_t^* = \underset{l=1,2,\ldots,c}{\arg\max} \ \hat{p}_t^l. \tag{13}$$

In summary, the proposed LSPD method extracts intrinsic spatial–spectral information from the limited labeled samples and their spatial–spectral neighbors, and it significantly enhances the discrimination capability through conducting spatial–spectral reconstruction on a multiclass probability map. The detailed procedure of LSPD is given in Algorithm 2.

According to Algorithm 2, we adopt big O notation to analyze the computational complexity of LSPD. The number of labeled samples is denoted as n. The process for selecting neighbors of labeled pixels costs $O(nw^2)$. The linear representation weight vectors $\{\mathbf{v}_t^l : l \in \{1, 2, \ldots, c\}\}$ of test samples are computed with the cost of $O(ck^3)$. The KNN dictionaries $\{\mathbf{K}_t^l : l \in \{1, 2, \ldots, c\}\}$ and the probabilities $\{\mathbf{p}_t^l : l \in \{1, 2, \ldots, c\}\}$ both take $O(c^2)$, and the cost of spatial–spectral reconstruction probabilistic vector $\boldsymbol{\beta}_t^S$ is $O(w^2)$. Then, the total computational complexity of LSPD is $O(nw^2 + ck^3 + c^2)$, which mainly depends on the number of neighbor points k and the size of spatial window w.

III. Experimental Results and Discussion

In this section, we perform the proposed LSPD method on some hyperspectral data sets and evaluate its classification performance by comparing with some state-of-the-art classifiers.

A. Data Sets

Indian Pines HSI was captured by Airborne Visible Infrared Imaging Spectrometer (AVIRIS) in 1992 over the northwest region of Indiana. The image consists of 145 × 145 pixels and 220 spectral bands in the range from 400 to 2400 nm. Due to noise or water absorption phenomenon, bands 104–108, 150–163, and 220 were removed from this data set. The geometric resolution is up to 20 m, and there are total 16 land-cover types considered in this image. The HSI in false color and its corresponding ground truth are shown in Fig. 4.

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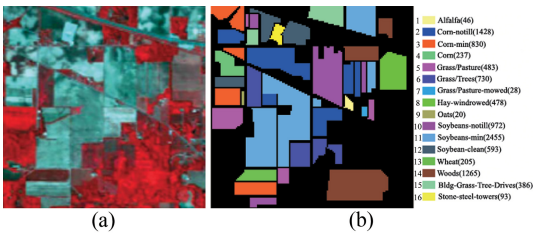


Fig. 4. Indian Pines HSI. (a) HSI in false color. (b) Ground truth.

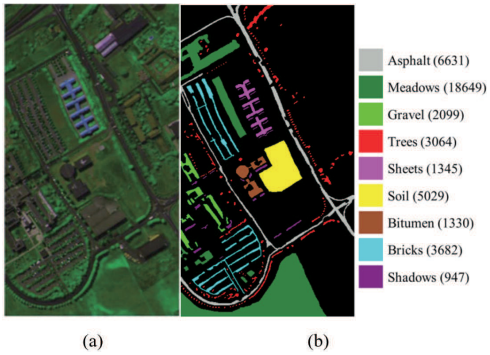


Fig. 5. PaviaU HSI. (a) HSI in false color. (b) Ground truth.

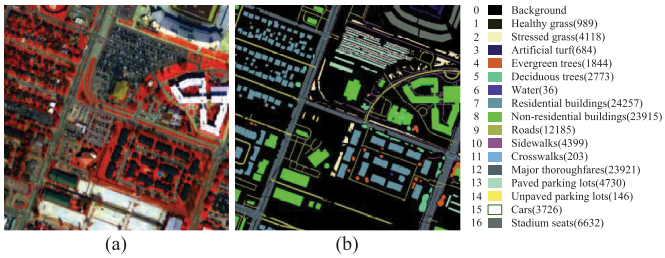


Fig. 6. HoustonU HSI. (a) HSI in false color. (b) Ground truth.

PaviaU HSI was taken over the urban area of Pavia University by the Reflective Optics System Imaging Spectrometer (ROSIS) in 2002. The image possesses 610×340 pixels with 1.3-m spatial resolution and 115 spectral bands covering the spectral range from 430 to 860 nm. After removing noise and water absorption bands, 103 channels have remained for experiments. Nine land-cover types are considered in this image. The false-color image and the ground truth map of PaviaU are shown in Fig. 5.

HoustonU HSI as a part of the hyperspectral data set used for the 2018 IEEE GRSS data fusion contest was acquired by the National Center for Airborne Laser Mapping (NCALM) over the University of Houston campus in 2017. It covers a 380- to 1050-nm spectral range with 48 bands at a 1-m ground sample distance. The spatial size of HoustonU image is up to 601×601 pixels, and total 16 classes of ground objects are considered in this image. The image in false color and its corresponding ground truth are shown in Fig. 6.

B. Experimental Setup

In the experiment, each HSI data set was randomly divided into training and test sets. The training set was applied to train a classification model, and then, all test samples were classified by the trained classifier. After that, the overall classification accuracy (OA), average classification accuracy (AA), and kappa coefficient (KC) were exploited to evaluate the performance of different classifiers.

On each hyperspectral data set, the proposed LSPD is compared with KNN [5], SVM [7], KNNS [11], DBN [32], SSDBN [35], JCR [27], SMLR-SpATV [31], IPDKNN [18], SOMP [26], SVMCK [25], SRMRF [28], SVMMRF [29], and aICPssi [30]. The former four methods are spectral-based classifiers, while the latter nine approaches all make use of spatial–spectral combined information for classification. For convenience, we simplify SMLR-SpATV as SMTV. In addition, we remove the multiclass probability map reconstruction from LSPD to construct a new classifier called LSPD no reconstruction (LSPDnr) and compare it with LSPD to evaluate the effectiveness of multiclass probability map reconstruction for HSI classification.

To achieve better results, we adopt cross validation to obtain the optimal parameters for different classifiers. For KNN and KNNS, the numbers of nearest neighbors are set to 1 and 7, respectively. The upper bounds on sparsity levels of SOMP and SMTV are given as 10. For DBN and SSDBN, their neural networks are constructed as frameworks with 103-80-80-9 units on the PaviaU data set and 200-110-110-16 units on the Indian Pines and HoustonU data sets, respectively. These networks are turned with 1000 epochs of pretraining and 5000 epochs of fine-tuning in experiments. In SMTV, the RBF kernel parameter σ and the smoothness parameter μ are set to 0.65 and 2, respectively. For the MRF-based methods SRMRF, SVMMRF, and aICPssi, the sizes of spatial neighborhood are set to 3×3 on the Indian Pines data set, and 5×5 on the PaviaU and HoustonU data sets. The smoothness parameter μ of SRMRF is set to 2 on the Indian Pines data set and 10 on the PaviaU and HoustonU data sets. The smoothness parameter μ of aICPssi is set to 10 on three data sets. The sizes of spatial windows in JCR, IPD-KNN, SSDBN, and SOMP are set to 9×9 for the Indian Pines data set and 5×5 for both the PaviaU and HoustonU data sets. The spatial window of SVMCK is selected as 13×13 for the Indian Pines data set and 9×9 for the PaviaU and HoustonU data sets. For LSPD, the optimized parameters w, k, and α are introduced in Section III-C. All experiments were performed on a personal computer with i5-4590 central processing unit, 8-G memory, and 64-bit Windows 10 using MATLAB 2014a.

C. Parameters Tuning

In the proposed LSPD, w, k, and α are three important parameters. The parameters w and k, respectively, determine the sizes of spatial window and spectral neighborhood, while the parameter α controls the weight of spatial–spectral reconstruction on a multiclass probability map.

To select the suitable parameters for three HSI data sets, we randomly chose 1% samples per class on the Indian Pines data set, 0.1% samples per class on the PaviaU data set, and 0.8% samples per class on the HoustonU data set for training. The parameters w and k were turned with the

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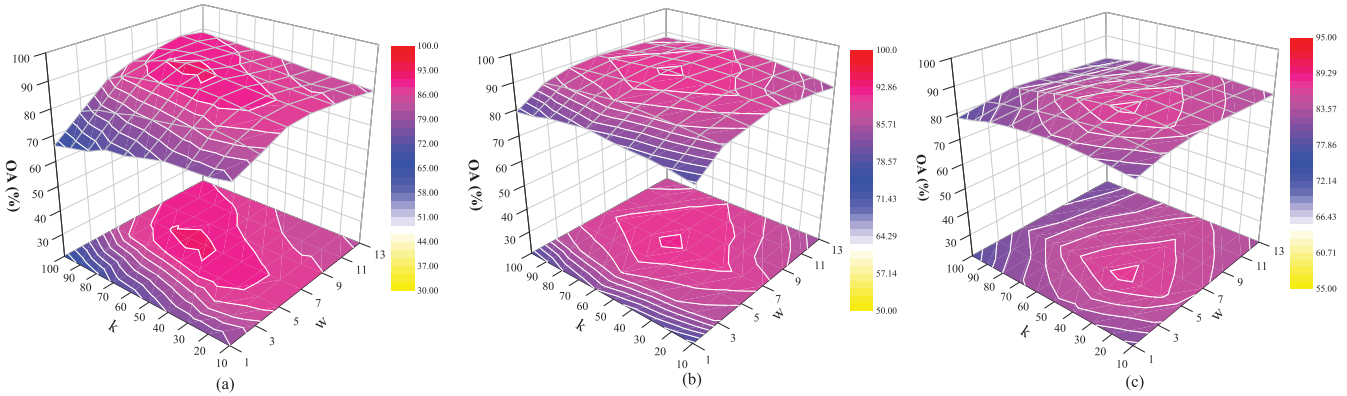


Fig. 7. Classification accuracy of LSPD with respect to parameters w and k. (a) Indian Pines. (b) PaviaU. (c) HoustonU.

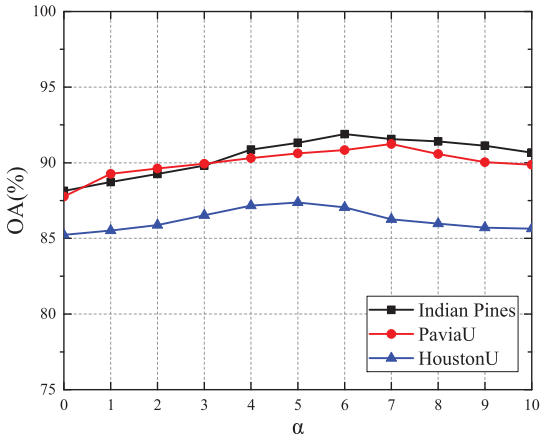


Fig. 8. Classification accuracy of LSPD with respect to parameter α.

sets of $\{1, 3, 5, \ldots, 13\}$ and $\{10, 20, 30, \ldots, 100\}$, respectively. While the tradeoff parameter α was given with a set $\{1, 2, 3, \ldots, 10\}$. Experiments were repeated ten times in each condition to reduce random errors. Fig. 7 shows the mean OA with respect to parameters w and k, and Fig. 8 shows the relationship between mean OA and parameter α.

According to Fig. 7, these curved surface maps indicate that the OAs fluctuate with the changes of w and k on three data sets. For parameter w, it displays a significant change in terms of classification accuracy with a fixed value of k, especially on the Indian Pines data set. With the increase of w, the OAs quickly reach to a peak value and then slightly decline for the reason that a larger spatial window containing more spatial neighbors is beneficial for HSI classification. However, if the size of the spatial window is too large, the neighbors will be more likely to come from different classes, which will bring negative influence for classification. The OAs slightly increase as the increasing of parameter k with a fixed value of w and then descend subtly when k keeps growing. This reason is that too many neighbors may increase the difference in the neighbor set and produce larger linear representation errors. According to the peak values generated in different positions on three curved surface maps, we fix w and k to be 7 and 70 for the Indian Pines data set, 7 and 60 for the PaviaU data set, 5 and 40 for the HoustonU data set.

For parameter α, it can be seen from Fig. 8 that the OAs on three HSI data sets first obviously improve and then mildly decrease with the increase of α. A too small or too large α value cannot effectively balance the proportion of spatial and spectral information in spatial–spectral weighted reconstruction. In order to achieve satisfying classification performance, α is set to 6 on the Indian Pines data set, 7 for the PaviaU data set, and 5 for the HoustonU data set.

D. Classification Performance Analysis

To investigate the classification performance of LSPD with different numbers of labeled training samples, we randomly chose a certain percentage of samples from each class for training, and the other samples were used to test the effectiveness of different classifiers. For the Indian Pines data set, we selected P ($P = 1\%, 2\%, 4\%, 6\%, 8\%$) samples per class for training. Due to the PaviaU and HoustonU data sets that possess more samples, the training proportion P is set as 1‰, 2‰, 4‰, 6‰, and 8‰, respectively. Each experiment was repeated 30 times to reduce the influence of random errors. Besides, we use the T-test to evaluate the statistical significance of LSPD. The repeated results of LSPD and other methods have been deal with two-sided T-test, and the confidence value of T-test is set to 0.05. The value of hypothesis $h = 1$ and the probability of null hypothesis $p < 0.05$ represent that the results of LSPD are significantly different with other methods; otherwise, the difference between two algorithms is not obvious. The values of p, h, and the mean of OAs with std (in percentage) on three HSI data sets are given in Tables I–III.

As shown in Tables I–III, the OA of each method is improved with the increase of training set for containing more priori information. It is obvious that all algorithms represent much better performance on the Indian Pines and PaviaU data sets than the HoustonU data set because HoustonU possesses more complex ground object distribution and less spectral bands than two other HSI data sets. The spatial–spectral combined classifiers, such as SSDBN, KNNSwr, JCR, SMTV, IPDKNN, SOMP, SVMCK, SEMRF, aICPssi, LSPDnr, and LSPD, are significantly better than the spectral-based classifiers, such as KNN, SVM, KNNS, and DBN, which denotes that the spatial information in HSI can help to enhance classification performance. DBN and SSDBN achieve unsatisfying

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TABLE I

CLASSIFICATION RESULTS WITH DIFFERENT NUMBERS OF TRAINING SAMPLES ON THE INDIAN PINES DATA SET [OA $\pm$ STD (p, h)]

Classifier	$P = 1\%$	$P = 2\%$	$P = 4\%$	$P = 6\%$	$P = 8\%$
KNN	51.05±1.72(8.5E-22, 1)	57.17±0.96(6.2E-23, 1)	62.41±1.37(1.5E-23, 1)	65.03±1.01(1.3E-29, 1)	66.81±0.51(1.4E-28, 1)
SVM	57.37±3.16(1.8E-16, 1)	65.03±1.27(1.1E-21, 1)	72.83±1.02(4.0E-21, 1)	76.69±0.69(5.7E-24, 1)	78.84±0.95(4.2E-26, 1)
KNNS	63.18±1.29(3.7E-19, 1)	65.91±0.83(1.5E-20, 1)	71.33±0.74(1.6E-24, 1)	73.83±0.39(7.9E-27, 1)	75.37±0.29(2.3E-30, 1)
DBN	54.18±2.31(2.2E-19, 1)	63.55±1.71(2.2E-19, 1)	69.74±0.94(1.3E-21, 1)	71.71±0.96(6.5E-26, 1)	71.67±0.81(1.7E-27, 1)
SSDBN	79.54±2.37(4.1E-09, 1)	81.98±2.04(2.2E-09, 1)	86.96±1.91(4.2E-12, 1)	89.38±1.02(4.1E-14, 1)	90.33±0.99(2.4E-14, 1)
IPDKNN	82.13±1.55(1.2E-10, 1)	86.55±1.05(2.6E-11, 1)	93.15±0.89(1.1E-08, 1)	94.81±0.66(1.6E-10, 1)	95.32±0.72(9.6E-16, 1)
SOMP	80.29±1.91(5.6E-12, 1)	85.28±0.99(1.0E-12, 1)	91.18±1.13(1.2E-12, 1)	94.25±0.81(1.2E-13, 1)	96.12±0.56(3.1E-11, 1)
SVMCK	84.98±1.92(3.7E-07, 1)	87.99±1.17(3.1E-10, 1)	93.9±0.63(1.15E-11, 1)	96.18±0.48(3.9E-09, 1)	97.44±0.51(5.2E-11, 1)
JCR	78.73±2.36(4.6E-13, 1)	84.68±1.33(2.1E-12, 1)	89.22±1.16(3.8E-13, 1)	92.47±0.55(1.6E-14, 1)	94.46±0.55(1.6E-11, 1)
SMTV	78.41±3.31(3.1E-09, 1)	85.55±1.82(2.2E-12, 1)	89.68±1.42(1.1E-11, 1)	93.71±0.93(9.1E-11, 1)	96.96±0.47(3.9E-09, 1)
SRMRF	79.74±2.04(4.1E-13, 1)	83.86±1.56(1.3E-11, 1)	90.14±0.95(1.8E-14, 1)	92.42±0.75(1.3E-13, 1)	94.74±0.71(1.9E-13, 1)
SVMMRF	75.39±2.52(1.3E-13, 1)	86.54±1.93(1.7E-10, 1)	90.59±1.81(3.4E-08, 1)	93.24±1.02(5.3E-11, 1)	94.52±0.96(4.2E-12, 1)
aICPssi	82.29±2.94(1.1E-09, 1)	86.25±2.58(8.7E-07, 1)	90.17±1.88(1.5E-08, 1)	92.41±1.54(3.3E-10, 1)	93.55±1.39(2.8E-12, 1)
LSPDnr	84.21±2.74(3.8E-07, 1)	87.34±1.24(9.9E-08, 1)	92.32±1.16(1.1E-11, 1)	95.72±0.89(8.9E-08, 1)	96.63±0.56(5.4E-06, 1)
LSPD	**90.79±1.74**	**94.75±1.72**	**97.13±0.53**	**98.43±0.52**	**98.81±0.36**

Note: The best results of a column are marked in bold, and xEy represents $x \times 10^y$.

TABLE II

CLASSIFICATION RESULTS WITH DIFFERENT NUMBERS OF TRAINING SAMPLES ON THE PAVIAU DATA SET [OA $\pm$ STD (p, h)]

Classifier	$P = 1‰$	$P = 2‰$	$P = 4‰$	$P = 6‰$	$P = 8‰$
KNN	70.33±2.17(8.7E-15, 1)	73.04±1.41(8.3E-18, 1)	76.62±0.69(2.2E-25, 1)	78.72±0.79(7.6E-24, 1)	79.25±0.68(5.1E-23, 1)
SVM	73.89±2.75(2.7E-13, 1)	79.02±1.88(1.7E-11, 1)	84.67±1.07(1.1E-17, 1)	88.36±0.97(1.8E-17, 1)	89.12±0.53(1.7E-20, 1)
KNNS	70.01±1.72(7.4E-15, 1)	71.47±1.69(2.1E-17, 1)	75.12±1.37(6.2E-22, 1)	76.89±0.92(1.7E-23, 1)	77.54±0.58(1.1E-26, 1)
DBN	69.67±3.30(1.0E-11, 1)	74.74±2.21(4.5E-15, 1)	76.85±1.52(2.9E-18, 1)	78.21±0.75(2.7E-24, 1)	79.81±0.57(2.3E-27, 1)
SSDBN	79.36±3.38(8.7E-10, 1)	83.97±2.61(2.4E-10, 1)	87.46±2.04(1.2E-11, 1)	88.29±1.02(1.6E-17, 1)	92.54±0.52(1.5E-15, 1)
IPDKNN	82.25±2.42(4.5E-06, 1)	88.13±1.25(6.5E-08, 1)	91.63±0.75(5.9E-12, 1)	93.31±0.54(4.6E-13, 1)	94.81±0.57(2.8E-11, 1)
SOMP	67.99±2.54(2.4E-13, 1)	69.17±2.03(6.2E-18, 1)	70.47±1.21(6.6E-23, 1)	72.57±1.08(7.8E-24, 1)	72.55±0.64(4.9E-29, 1)
SVMCK	87.03±2.79(2.6E-03, 1)	89.29±1.90(1.4E-05, 1)	93.17±0.92(3.8E-09, 1)	96.34±0.75(2.1E-06, 1)	96.36±0.68(4.1E-07, 1)
JCR	82.98±3.75(5.1E-05, 1)	84.06±2.34(4.9E-08, 1)	88.94±2.26(4.5E-07, 1)	91.34±1.19(1.2E-10, 1)	92.37±0.66(1.2E-16, 1)
SMTV	72.91±2.73(8.6E-14, 1)	77.24±1.59(2.3E-16, 1)	80.11±1.26(1.6E-17, 1)	83.56±0.62(3.3E-24, 1)	86.74±0.32(1.2E-24, 1)
SRMRF	81.04±1.18(2.5E-08, 1)	85.32±0.88(1.6E-12, 1)	88.93±0.95(5.2E-13, 1)	91.66±0.42(2.1E-18, 1)	93.05±0.82(5.3E-16, 1)
SVMMRF	80.39±3.59(2.3E-04, 1)	83.61±2.27(3.6E-11, 1)	90.86±1.54(1.5E-11, 1)	94.74±1.04(2.7E-10, 1)	96.13±0.52(6.2E-11, 1)
aICPssi	87.59±2.22(1.1E-01, 0)	91.71±1.96(7.3E-02, 0)	94.46±1.35(4.0E-03, 1)	96.51±0.77(1.7E-05, 1)	97.24±0.59(2.1E-04, 1)
LSPDnr	84.25±2.19(5.8E-05, 1)	89.08±1.37(5.1E-08, 1)	91.25±0.72(6.5E-13, 1)	91.68±0.68(1.2E-14, 1)	92.52±0.49(4.8E-17, 1)
LSPD	**90.4±2.68**	**93.63±1.68**	**96.52±0.59**	**97.77±0.39**	**98.5±0.36**

Note: The best results of a column are marked in bold, and xEy represents $x \times 10^y$.

TABLE III

CLASSIFICATION RESULTS WITH DIFFERENT NUMBERS OF TRAINING SAMPLES ON THE HOUSTONU DATA SET [OA $\pm$ STD (p, h)]

Classifier	$P = 1‰$	$P = 2‰$	$P = 4‰$	$P = 6‰$	$P = 8‰$
KNN	49.71±1.72(8.5E-16, 1)	56.12±1.02(5.8E-23, 1)	59.66±0.93(1.5E-20, 1)	62.55±0.84(3.9E-23, 1)	63.37±0.66(8.2E-29, 1)
SVM	61.27±1.27(1.4E-16, 1)	69.02±0.89(8.1E-16, 1)	71.98±0.62(3.6E-16, 1)	74.48±0.44(6.0E-23, 1)	75.79±0.26(1.8E-25, 1)
KNNS	63.88±2.04(6.6E-10, 1)	67.93±1.69(4.0E-13, 1)	73.89±1.04(3.8E-17, 1)	76.03±0.57(3.5E-21, 1)	77.74±0.52(8.2E-20, 1)
DBN	56.46±3.56(3.9E-12, 1)	65.14±2.28(2.1E-14, 1)	69.05±2.04(1.8E-13, 1)	71.48±0.73(2.1E-22, 1)	71.49±0.71(1.3E-21, 1)
SSDBN	58.14±4.63(1.9E-09, 1)	66.89±2.58(7.6E-13, 1)	71.27±2.45(1.1E-11, 1)	74.43±2.10(2.5E-14, 1)	74.92±2.06(2.8E-11, 1)
IPDKNN	55.04±1.99(4.4E-15, 1)	61.14±1.22(6.9E-22, 1)	65.97±0.95(1.1E-19, 1)	69.34±0.45(3.5E-27, 1)	71.17±0.31(9.8E-29, 1)
SOMP	66.39±2.49(2.5E-09, 1)	70.81±1.17(1.2E-14, 1)	73.91±1.05(3.0E-13, 1)	75.57±0.63(3.5E-21, 1)	75.99±0.43(1.0E-24, 1)
SVMCK	65.93±0.93(3.4E-11, 1)	72.83±0.72(3.3E-14, 1)	78.27±0.53(1.1E-12, 1)	81.07±0.42(1.8E-17, 1)	83.67±0.25(2.4E-15, 1)
JCR	63.66±1.13(6.2E-14, 1)	72.85±0.72(5.4E-18, 1)	74.58±0.68(1.2E-14, 1)	76.98±0.44(2.6E-19, 1)	78.65±0.38(2.1E-19, 1)
SMTV	63.32±1.37(1.2E-13, 1)	70.64±1.21(4.1E-14, 1)	74.32±0.97(6.8E-13, 1)	77.94±0.42(1.1E-18, 1)	79.33±0.35(1.9E-19, 1)
SRMRF	74.31±1.93(1.9E-01, 0)	79.21±0.41(1.0E-02, 1)	79.75±0.39(4.3E-09, 1)	81.16±0.22(3.7E-15, 1)	82.37±0.17(5.8E-23, 1)
SVMMRF	69.66±4.35(2.3E-08, 1)	76.88±1.98(3.3E-07, 1)	79.42±0.87(2.6E-08, 1)	83.81±0.68(2.2E-08, 1)	85.35±0.63(1.2E-09, 1)
aICPssi	70.02±3.21(4.0E-02, 1)	75.83±1.32(1.2E-10, 1)	79.58±0.63(1.3E-10, 1)	82.48±0.31(3.2E-14, 1)	84.54±0.12(1.2E-16, 1)
LSPDnr	73.59±1.65(1.8E-02, 1)	78.27±0.72(3.6E-07, 1)	80.41±0.41(3.7E-10, 1)	83.37±0.25(1.8E-12, 1)	83.31±0.21(1.4E-19, 1)
LSPD	**75.26±1.85**	**81.02±0.95**	**83.39±0.91**	**86.17±0.4**	**87.21±0.31**

Note: The best results of a column are marked in bold, and xEy represents $x \times 10^y$.

classification results for lacking enough samples to train their deep networks. In many studies, deep network-based classifiers usually choose half or more samples for training to achieve satisfying classification performance. But this limits the application of deep networks in real HSI classification.

For SRMRF and aICPssi, they achieve good classification accuracies compared with many methods. SRMRF and aICPssi as MRF-based methods not only construct a probabilistic model using *a priori* spectral information but also build a multilevel logistic (MLL) model to fuse spatial information in classifiers. Although they both construct probabilistic models, there are two major differences between these two MRF-based methods and the proposed LSPD. One is that SRMRF and aICPssi utilize SRC or SVM to preclassification for obtaining the class probabilities of samples, while LSPD computes the class probabilities via calculating the local linear reconstruction errors of samples on each class. The other one is that SRMRF and aICPssi obtain labels by solving the optimization function of MRF with the maximum *a posterior* (MAP) decision rule, but LSPD constructs the multiclass probability map and weighted reconstructs this map to obtain the spatial–spectral probabilistic distribution for predicting

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the labels of samples. Therefore, LSPD fully considers the geometric structure in the local spectral neighborhood and the spatial information in HSIs, which results in LSPD presenting better classification performance than SRMRF and aICPssi.

Compared with other spatial–spectral combined classifiers, the proposed LSPD method achieves the best results on three HSI data sets, especially under small training set. On the Indian Pines and PaviaU data sets, the classification accuracies of LSPD are higher than 90% when only 1% or 0.1% samples are selected for training. On the HoustonU data set, LSPD still produces the highest classification accuracy among all algorithms. The reason for this phenomenon is that LSPD exploits the spatial–spectral neighbors of limited labeled samples to help training classification model, and this strategy can significantly improve the classification performance when only a few labeled training samples can be available. To further demonstrate the effectiveness of this strategy, we remove the multiclass probability map from LSPD and perform the modified method LSPDnr on HSI data sets. Although LSPDnr lacks the step of multiclass probability map reconstruction, it is superior to most of the classification methods on three data sets. Thus, the spatial–spectral neighbors of labeled samples can provide helpful discriminant spatial–spectral information to improve the classification performance.

To explore the classification accuracy for each land-cover type on three HSI data sets, we randomly selected 1% samples per class on the Indian Pines data set, 0.2% samples per class on the PaviaU data set, and 0.8% samples per class on the HoustonU data set as training sets, respectively. The remaining samples were tested by different classification algorithms. For the classes containing a very small number of samples, we took ten or five samples from each class for training. Tables IV–VI show the classification accuracies of each class with different classifiers on three HSI data sets, and the corresponding classification maps are shown in Figs. 9–11.

From Tables IV–VI, we can see that spatial–spectral combined classifiers give better classification results than the spectral-based classifiers in most of the classes, which indicates that spatial information is helpful to improve classification performance. Among all spatial–spectral combined classifiers, the proposed LSPD method produces the best classification results on three HSI data sets. It achieves the best classification accuracies for 12 classes on the Indian Pines data set, 6 classes on the PaviaU data set, and 10 classes on the HoustonU data set. LSPD considers the geometric structure in a spectral neighborhood by local linear representation, so it can classify the samples from similar land-cover types effectively. Furthermore, LSPD exploits the spatial–spectral neighbors of limited labeled samples for classification, which is conducive to improving the classification accuracy for classes with small-scale samples. In all methods, LSPD spends the most computing time because it selects spatial–spectral neighbors of labeled samples and adds these neighbors into spatial–spectral training set for classification. More training samples must cost more time in program running. Furthermore, calculating the local weighted reconstruction of samples in all classes is also a major reason for high time-consuming of LSPD.

As the classification maps shown in Figs. 9–11, it is clear that the spatial–spectral combined classifiers generate more homogenous areas than the traditional spectral-based classifiers. The proposed LSPD method produces smoother classification maps compared with other classifiers, especially in the regions of *Corn-notill (class2), Cornmin (class3), Soybeans-notill (class10), and Soybeans-clean (class12)* for the Indian Pines data set, *Soil (class6), Gravel (class9), and Meadows (class2)* for the PaviaU data set, and *Evergreen trees (class4), Deciduous trees (class5), and Cars (class15)* for the HoustonU data set. All classification results demonstrate that LSPD is more effective than many spatial–spectral-based approaches to produce fewer misclassification pixels because it exploits the local linear reconstruction and the multiclass probability map to reduce the differences between pixels in the homogeneous areas.

E. State-of-the-Art Comparison

To further verify the effectiveness of the proposed method, we compare LSPD with several state-of-the art algorithms, such as RF [8], MICA-SVM [21], SCAE-SVM [21], DAFE [36], GLCM [22], EMPA [20], EPF [23], IFRF [24], and Gaussian mixture model (GMM) [38]. GMM generates some synthetic data points by sampling from the filtered features extracted from HSI, and the synthetic data points are also adopted for training. GMM is combined with the spatial–spectral feature extraction methods EMAP-PCA and EMAP-NWFE to form two classifiers termed EMAP-PCA-GMM and EMAP-NWFE-GMM [38].

In addition, some deep learning methods, including CNN [33], SSAE [34], SuSA [36], capsule network (CapsNet) [39], multigrained network (MugNet) [40], and rolling vertex component analysis network (RVCANet) [41] are also exploited for comparison. The modified two-layer CapsNets are inspired by the simplicity of shallower deep learning models, and it significantly improves the classification performance with limited training samples. MugNet and RVCANet as two modified deep learning algorithms are presented to extract the spectral–spatial features via a multigrained scanning strategy or a rolling guidance filter, respectively. Then, MugNet and RVCANet process the extracted spatial–spectral features with their modified networks for classification.

Furthermore, the Salinas data set is also used in the following experiments. In experiments, N training samples were randomly selected from each class, and the remaining samples were used as test samples. The results of the previously existing algorithms are directly obtained from the corresponding studies, while the results of the proposed LSPD are achieved by performing experiments under the same training conditions. The experimental results on three HSI data sets are given in Tables VII–IX.

As shown in Tables VII–IX, it can be clearly seen that the fitter-based methods and the feature extraction-based classifiers, including EPF, IFRF, EMAP, GLCM, and MICA-SVM, produce unsatisfying classification results in comparison with other spatial–spectral classifiers. The reason is that these traditional classifiers with spatial features are unable to fully consider the spatial correlation between the samples in a

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TABLE IV

CLASSIFICATION RESULTS OF DIFFERENT CLASSIFIERS ON THE INDIAN PINES DATA SET

Class	Samples		Classifiers														
	Train	Test	KNN	SVM	KNNS	DBN	SSDBN	SVMMRF	JCR	SMTV	IPDKNN	SOMP	SVMCK	SRMRF	aICPssi	LSPDnr	LSPD
1	10	56	83.3	83.3	80.6	80.8	97.2	91.7	94.4	97.2	**100**	**100**	94.4	**100**	97.2	**100**	97.2
2	14	1414	34.9	41.7	53.6	37.1	81.7	38.1	61.7	57.6	68.0	70.0	77.3	88.4	74.1	82.2	**90.8**
3	10	820	26.7	37.4	41.0	33.2	60.6	59.3	71.6	58.9	64.9	51.1	72.7	53.2	63.4	70.9	**88.7**
4	10	227	53.3	70.0	68.7	60.4	99.1	**100**	92.5	**100**	**100**	82.8	93.8	91.5	98.7	96.5	94.3
5	10	473	46.9	54.3	78.2	58.6	83.1	93.9	85.9	88.6	89.6	96.2	96.1	**99.2**	68.7	84.2	86.9
6	10	720	63.5	80.7	80.8	75.7	99.2	99.4	86.9	**99.6**	91.5	95.3	82.2	97.9	92.4	93.2	92.1
7	10	018	94.4	94.4	94.4	88.9	66.7	88.9	**100**	**100**	**100**	94.4	**100**	94.4	**100**	**100**	**100**
8	10	468	83.3	85.9	88.9	82.9	94.9	**100**	97.9	84.6	99.6	94.9	97.2	91	99.8	**100**	**100**
9	10	010	90.0	80.0	**100**	90.0	**100**	**100**	**100**	97.2	**100**	80.0	**100**	80.0	**100**	**100**	**100**
10	25	962	44.0	48.0	51.2	41.9	46.8	34.9	70.8	76.1	79.3	62.7	72.5	44.9	72.4	78.2	**88.4**
11	10	2430	51.3	65.3	58.9	57.6	83.2	90.5	80.2	89.2	87.4	93.2	80.0	77.2	85.6	82.7	**89.4**
12	10	583	30.0	32.4	34.8	27.4	63.6	55.1	49.6	30.7	62.8	60.9	65.4	68.6	64.5	75.6	**90.4**
13	10	195	73.9	96.4	98.5	77.4	99.5	99.5	98.5	98.9	85.6	**100**	99.5	**100**	96.4	96.4	**100**
14	13	1252	69.4	66.5	84.3	97.1	98.3	99.2	92.7	97.8	92.1	99.1	96.4	83.2	92.1	95.8	**99.6**
15	10	376	29.5	35.9	36.7	30.9	85.1	70.7	73.9	41.8	75.8	49.2	85.4	88.6	96.8	76.1	**97.3**
16	10	83	95.2	97.6	95.2	95.2	**100**	**100**	96.4	**100**	**100**	98.8	**100**	**100**	**100**	98.9	**100**
	OA (%)		50.7	57.9	62.2	54.6	80.3	75.4	78.7	78.3	82.3	81.0	85.4	78.4	82.0	84.7	**90.8**
	AA (%)		60.6	66.9	71.6	63.6	84.6	82.6	84.9	82.7	87.4	83.0	87.4	84.3	87.6	89.5	**93.4**
	KC (%)		0.43	0.52	0.57	0.49	0.77	0.71	0.75	0.75	0.79	0.78	0.78	0.75	0.79	0.79	**0.9**
	Time (s)		19	14	157	51	326	25	133	84	553	389	326	89	78	499	703

Note: The best results of a row are marked in bold.

TABLE V

CLASSIFICATION RESULTS OF DIFFERENT CLASSIFIERS ON THE PAVIAU DATA SET

Class	Samples		Classifiers														
	Train	Test	KNN	SVM	KNNS	DBN	SSDBN	SVMMRF	JCR	SMTV	IPDKNN	SOMP	SVMCK	SRMRF	aICPssi	LSPDnr	LSPD
1	13	6618	68.6	74.9	47.9	54.9	69.2	88.6	83.5	55.1	93.0	16.7	87.2	94.6	93.2	90.9	**95.2**
2	37	18612	86.6	97.7	90.3	95.5	96.3	97.3	98.9	96.6	98.0	99.2	95.7	89.7	98.2	95.2	**99.9**
3	5	2094	35.5	60.5	46.5	48.1	46.4	62.7	63.6	57.4	64.4	95.4	51.1	82.5	**99.0**	70.5	52.0
4	6	3058	57.3	56.2	58.3	55.7	88.3	81.5	75.7	88.1	62.7	**99.5**	96.2	86.9	89.3	91.5	94.7
5	5	1340	97.3	99.4	99.5	99.8	97.9	**100**	99.9	98.4	96.4	99.8	99.8	99.5	**100**	97.3	**100**
6	10	5019	35.5	40.8	28.6	47.4	61.7	37.6	58.9	28.3	76.7	12.0	80.5	**85.8**	64.4	78.2	84.4
7	5	1325	88.0	75.5	85.2	88.9	85.7	91.3	72.1	77.4	63.9	66.2	81.5	74.1	93.8	87.4	**99.6**
8	7	3675	69.5	69.9	69.4	81.9	83.6	78.1	79.5	77.5	84.7	26.4	87.3	18.7	69.5	66.7	**95.1**
9	5	942	99.9	88.4	95.1	99.7	99.9	**100**	89.4	87.2	64.7	95.7	94.1	99.8	99.0	99.4	**100**
	OA (%)		72.4	79.4	70.5	75.5	83.7	84.4	85.8	77.2	87.6	68.7	89.4	84.2	90.3	88.5	**93.4**
	AA (%)		70.9	73.7	69.1	73.7	80.9	81.9	80.2	74.1	78.3	67.9	85.9	81.2	89.6	86.3	**90.3**
	KC (%)		0.63	0.72	0.59	0.68	0.78	0.79	0.81	0.69	0.83	0.58	0.87	0.79	0.87	0.85	**0.92**
	Time (s)		43	26	431	112	760	73	412	202	1183	948	882	183	174	1185	1598

TABLE VI

CLASSIFICATION RESULTS OF DIFFERENT CLASSIFIERS ON THE HOUSTONU DATA SET

Class	Samples		Classifiers														
	Train	Test	KNN	SVM	KNNS	DBN	SSDBN	SVMMRF	JCR	SMTV	IPDKNN	SOMP	SVMCK	SRMRF	aICPssi	LSPDnr	LSPD
1	10	979	75.8	50.6	82.9	74.5	62.5	56.9	62.2	79.2	74.5	49.7	90.2	57.3	87.9	79.9	72
2	33	4085	64.6	88.4	84.6	82.6	88.3	88.5	84.9	82.4	82.6	91.9	85	91.5	89.6	79.4	**92.1**
3	10	674	99.9	**100**	**100**	99.9	99.6	**100**	98.5	89.3	99.9	**100**	99.4	**100**	**100**	99.4	**100**
4	15	1829	82.8	60.3	82.9	84.9	50.4	96.9	51.6	72.7	84.9	73.6	85.9	79.9	94	82.7	**100**
5	22	2751	61.7	66.6	66.6	76.8	49.3	80.9	61.5	66.6	76.8	64.6	84.4	47.4	77.3	65.4	**87.3**
6	10	26	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**
7	194	24063	76.8	90.2	91.8	72.6	90.7	97.8	89.5	91.6	72.6	87.8	94	97.6	94	94.3	**98.7**
8	191	23724	68.8	89.9	83.8	83.5	88	94.3	90.5	90.4	83.5	89.2	88.6	96.1	88	89.4	**96.5**
9	97	12088	36.7	**53.3**	36.1	43.5	52.3	35.5	49.5	46	43.5	52.9	51.7	19.4	44.6	53.6	40.2
10	35	4364	33.8	41.9	33	40.5	**54.8**	33.1	33.7	44.7	40.5	41.1	34.1	18.3	32.7	40.5	38.5
11	10	193	17.1	21.2	21.8	**38.3**	20.7	0.0	35.8	32.6	38.3	26.9	23.3	0.0	24.9	27.5	19.3
12	191	23730	64.7	69.5	85.4	70	62.8	98.4	80.7	82.1	70	70	88.1	**99.9**	96.8	90.6	98.1
13	38	4692	58.3	63.5	79	80.6	64.9	**88.1**	68.3	68.9	80.6	63	87.2	88.5	91.2	85.1	85.2
14	10	136	99.3	98.5	99.3	79.4	89.7	**100**	93.4	95.6	79.4	95.6	99.3	**100**	**100**	**100**	**100**
15	30	3696	28.7	35	49.2	45.7	51.9	64.5	69.5	54.9	45.7	56.4	78	40.6	67.6	84.3	**89.3**
16	53	6579	76.5	87.4	93.7	95.3	92.8	**100**	94.8	95.5	95.3	84.9	97.9	**100**	98.4	98.2	**100**
	OA (%)		63.6	75.6	77.7	71.4	74.6	85.5	78.7	79.4	71.2	75.7	83.6	82.1	84.3	84.1	**87.9**
	AA (%)		65.3	69.8	74.4	73.2	69.9	82.3	72.8	74.5	73.1	71.7	80.4	75.8	80.4	79.4	**86.5**
	KC (%)		0.57	0.71	0.73	0.67	0.70	0.83	0.74	0.75	0.67	0.71	0.8	0.78	0.81	0.81	**0.86**
	Time (s)		98	67	1049	243	1767	172	834	449	2668	1868	2219	382	377	2862	3579

classification process. EMAP-PCA-GMM and EMAP-NWFE-GMM represent better classification performance than EMAP due to the synthetic data points produced by GMM that provide more discriminant information for classification. The spatial–spectral deep learning methods SSAE, SuSA, MugNet, and CapsNet generally achieve good classification accuracies;

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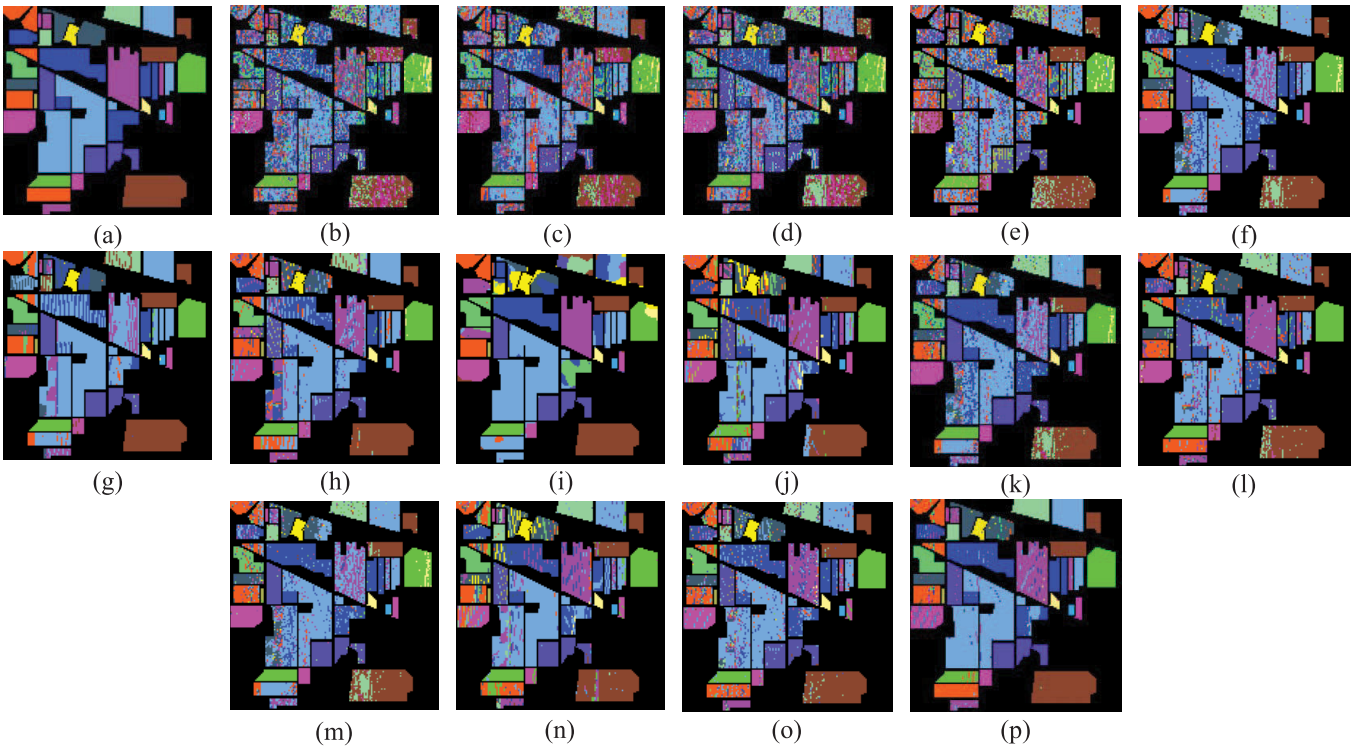


Fig. 9. Classification maps of different classifiers on the Indian Pines data set. (a) Ground truth, (b) KNN (49.7%), (c) SVM (57.9%), (d) KNNS (62.2%), (e) DBN (54.6%), (f) SSDBN (80.3%), (g) SVMMRF (75.4%), (h) JCR (78.7%), (i) SMTV (78.3%), (j) IPDKNN (82.3%), (k) SOMP (81.0%), (l) SVMCK (85.4%), (m) SRMRF (78.4%), (n) aICPssi (80.2%), (O) LSPDnr (84.7%), and (P) LSPD (90.8%).

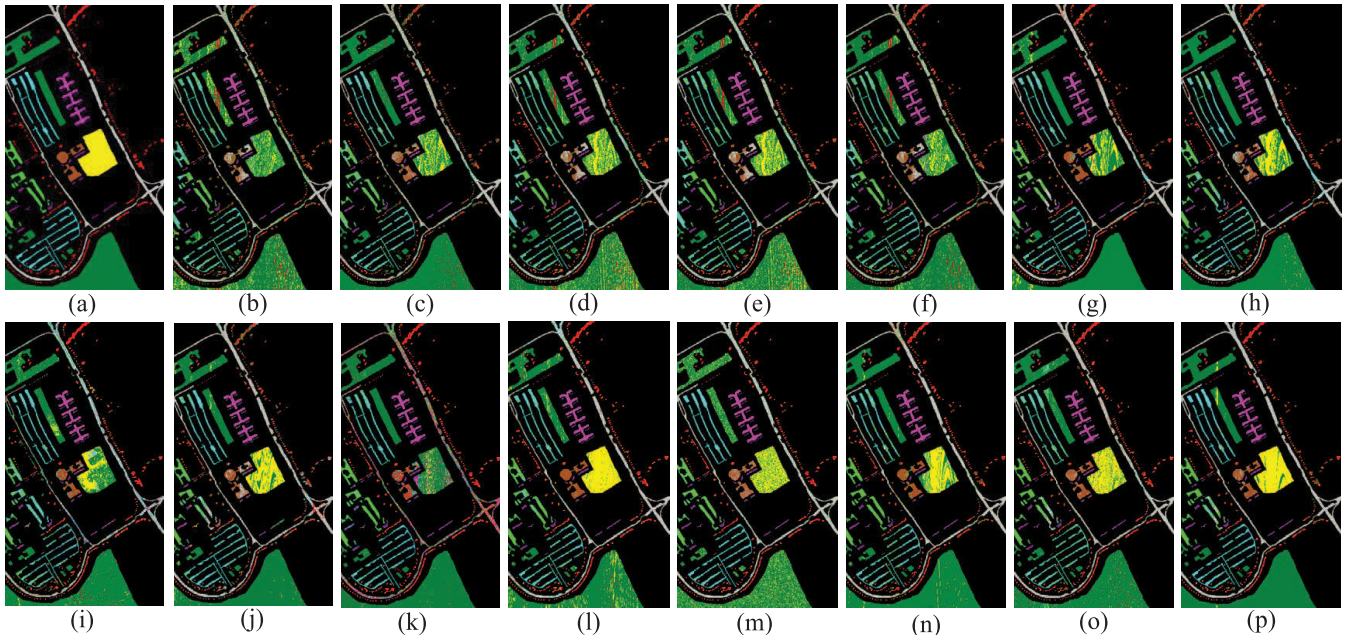


Fig. 10. Classification maps of different classifiers on the PaviaU data set. (a) Ground truth, (b) KNN (72.4%), (c) SVM (79.4%), (d) KNNS (70.5%), (e)DBN (75.5%), (f) SSDBN (83.7%), (g) SVMMRF (84.4%), (h) JCR (85.8%), (i) SMTV (77.2%), (j) IPDKNN (87.6%), (k) SOMP (68.7%), (l) SVMCK (89.4%), (m) SRMRF (84.2%), (n) aICPssi (90.3%), (o) LSPDnr (88.5%), and (p) LSPD (93.4%).

for the reason, they modify their deep networks based on the spectral and spatial characteristics in HSI to extract the spatial–spectral deep features for classifying unknown pixels.

On three HSI data sets, the proposed LSPD method outperforms other state-of-the-art classification algorithms in most training conditions. These classification results demonstrate that the local linear representation and spatial–spectral multiclass probability map reconstruction in LSPD are beneficial to enhance the discriminative power of land-cover classification. Furthermore, LSPD exhibits excellent classification

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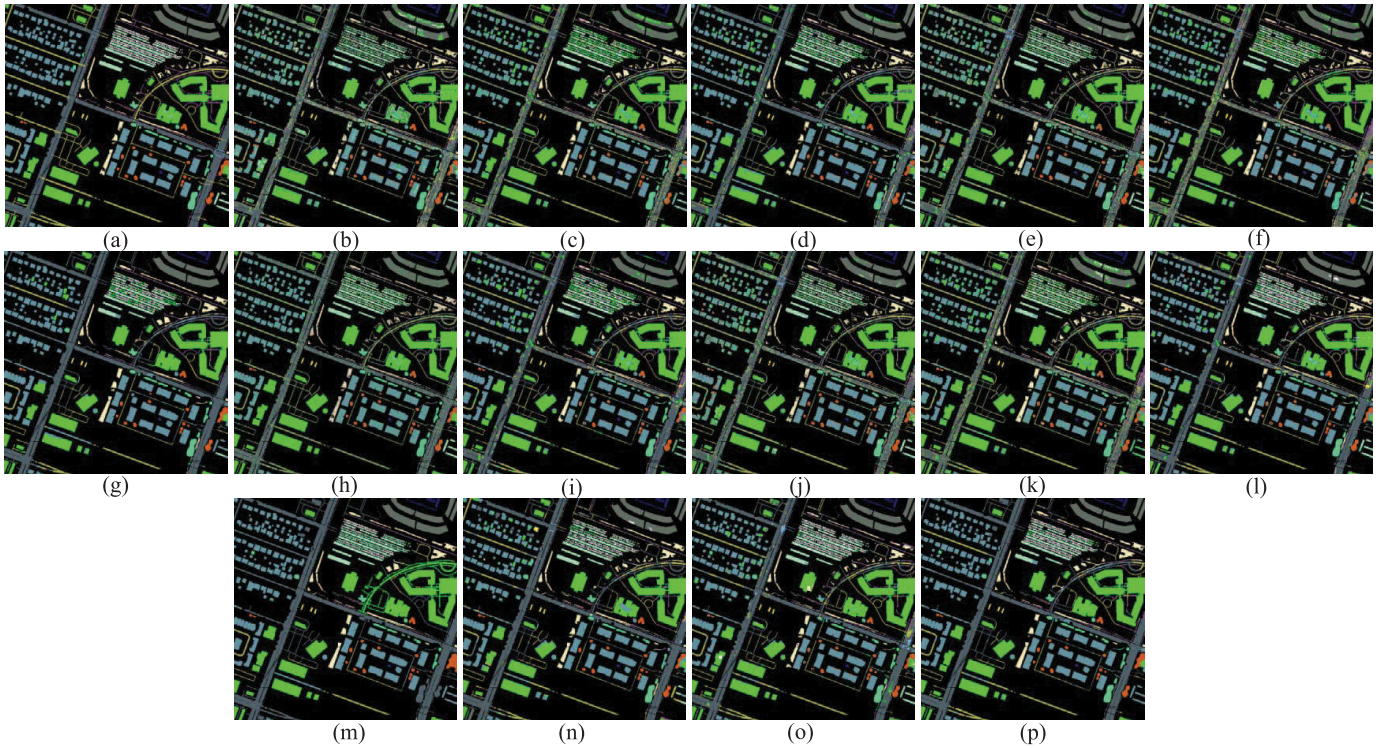


Fig. 11. Classification maps of different classifiers on the HoustonU data set. (a) Ground truth, (b) KNN (63.6%), (c) SVM (75.6%), (d) KNNS (77.7%), (e) DBN (71.4%), (f) SSDBN (74.6%), (g) SVMMRF (85.5%), (h) JCR (78.7%), (i) SMTV (79.4%), (j) IPDKNN (71.2%), (k) SOMP (75.7%), (l) SVMCK (83.6%), (m) SRMRF (82.1%), (n) aICPssi (84.3%), (o) LSPDnr (84.1%), and (p) LSPD (87.9%).

TABLE VII

COMPARISON OF LSPD AGAINST OTHER STATE-OF-THE-ART METHODS ON THE INDIAN PINES DATA SET [OA ± STD (%)]

Classifier	$N = 10$	$N = 20$	$N = 50$
MICA-SVM[36]	66.32±2.17	—	94.63±1.00
SCAE-SVM[36]	80.58±2.13	—	96.12±0.78
DAFE[36]	—	—	93.27
SuSA[36]	81.16±1.85	—	96.49±0.69
EPF[40]	—	77.54±3.10	—
IFRF[40]	—	88.85±2.51	—
RVCANet[40]	—	83.06±2.32	—
MagNet[40]	—	90.65±1.77	—
LSPD	**89.73±1.67**	**94.76±1.29**	**97.44±0.94**

Note: "—" represents the result is not given in the corresponding literature.

TABLE VIII

COMPARISON OF LSPD AGAINST OTHER STATE-OF-THE-ART METHODS ON THE PAVIAU DATA SET [OA ± STD (%)]

Classifier	$N = 10$	$N = 30$	$N = 50$	$N = 60$
EMAP[38]	—	70.94±3.55	—	—
EMAP-PCA-GMM[38]	—	83.05±2.29	—	—
EMAP-NWFE-GMM[38]	—	80.44±2.56	—	—
MICA-SVM[36]	75.74±3.81	—	93.92±1.38	—
SCAE-SVM[36]	84.67±3.36	—	95.84±0.94	—
SSAE[36]	—	—	91.96±0.87	—
SuSA[36]	89.28±2.64	—	**98.18±0.53**	—
RF[39]	—	—	—	71.89±1.24
CNN[39]	—	—	—	95.11±0.97
CapsNets[39]	—	—	—	95.90±1.38
LSPD	**90.19±2.15**	**95.23±1.68**	97.82±0.76	**98.31±0.52**

Note: "—" represents the result is not given in the corresponding literature.

TABLE IX

COMPARISON OF LSPD AGAINST OTHER STATE-OF-THE-ART METHODS ON THE SALINAS DATA SET [OA ± STD (%)]

Classifier	$N = 10$	$N = 30$	$N = 40$	$N = 50$
EMAP[38]	—	79.74±2.56	84.52±1.76	—
EMAP-PCA-GMM[38]	—	85.38±1.40	86.59±0.85	—
EMAP-NWFE-GMM[38]	—	88.28±1.01	89.17±0.81	—
MICA-SVM[36]	90.08±1.50	—	—	97.15±0.56
SCAE-SVM[36]	92.74±1.42	—	—	98.06±0.45
GLCM[36]	—	—	—	95.41
SuSA[36]	**93.47±1.27**	—	—	98.10±0.61
LSPD	92.28±1.36	**93.39±1.17**	**96.72±0.59**	**98.49±0.83**

Note: "—" represents the result is not given in the corresponding literature.

performance under small training condition because it takes full use of the spatial relationship between the pixels to select spatial–spectral neighbors for a supplementing training set, which is helpful to exploit more discriminant spatial–spectral information for classification.

IV. CONCLUSION

In this article, we propose a new classification method called LSPD based on the local geometric structure and spatial correlation in HSI. The proposed method first designs a new spatial–spectral neighbor selection model to capture more discriminant spatial–spectral information for training, which significantly improves the identification ability of the classifier with small labeled training samples. Based on the geometric structural characteristics of HSI data, LSPD calculates the class probabilities of test samples via minimizing local linear representation errors. With the class probabilities and pixels coordinates, a multiclass probability map is built for spatial–spectral weighted reconstruction to obtain the spatial–spectral

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probabilistic distribution of samples. Finally, the labels of test samples are predicted according to the spatial–spectral probabilities in different classes.

The experimental results on the Indian Pines, PaviaU, HoustonU, and Salinas hyperspectral data sets show that the proposed LSPD method achieves better classification results in comparison with many state-of-the-art classification methods. Our future work will focus on exploiting the multiple features in the HSI data to further enhance the classification performance.

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